Kalvin Kayi Mwashi

Software Engineer (ML & Full-Stack)
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GitHub | Portfolio | LinkedIn

Professional Summary

A versatile Software Engineer with proven experience building end-to-end web applications and AI-powered features. Combines strong full-stack skills (React, Node.js, Python, MongoDB, PostgreSQL) with practical machine learning knowledge to deliver scalable, user-focused solutions. Has integrated mobile payments (M-Pesa) and built production healthcare and e-commerce platforms. Passionate about clean code, system design, and solving real-world problems in Kenya and globally.

Areas of Expertise

- Full Stack Web Development (MERN, Flask, REST APIs)
- Database Design (SQL, NoSQL)
- Third-Party API Integration (M-Pesa Daraja, Payment Callbacks)
- Machine Learning & NLP (Scikit-learn, Hugging Face)
- Agile & Collaborative Development (Git, GitHub, Project Planning)

Technical Skills

- **Languages:** Python, JavaScript, SQL, HTML5, CSS3
- **Frontend:** React, Vite, Tailwind CSS, Framer Motion, shadcn/ui
- **Backend:** Node.js/Express, Flask, RESTful API design
- **Databases & Messaging:** MongoDB, PostgreSQL, Redis
- **ML & AI:** Scikit-learn, TensorFlow, Hugging Face Transformers, Pandas, NumPy, spaCy
- **DevOps & Tools:** Docker, Docker Compose, Nginx, Vercel, Git, GitHub, Power BI

Experience

Full Stack Developer | Anchored Healthcare (Nairobi, Kenya)
August 2025 – December 2025

- Architected and deployed a Proof of Concept (PoC) platform to connect healthcare professionals with available facilities in Kenya, focusing on streamlined user acquisition and responsive design.
- Engineered a fully responsive React frontend optimized for mobile-first engagement ensuring that busy clinicians could browse and apply for opportunities from any device—a key requirement for user experience in a healthcare setting.
- Designed a secure RESTful API (Node.js/Express) to handle professional profiles and facility matching logic, demonstrating proficiency in systems integration and secure data handling in a healthcare context.
- Managed production deployment on a self-managed Virtual Private Server (VPS) , configuring Nginx as a reverse proxy to serve the optimized React build (npm run build) and managing the Node.js backend process using PM2 for high availability, automatic restarts, and log management.
- Ensured system reliability and uptime in a hybrid-like environment by implementing PM2 process monitoring and basic firewall rules, demonstrating practical DevOps and infrastructure management skills.

Full Stack Developer & Digital Literacy Lead | DigiMagic Tech

(in partnership with Liberating Education e.V.) | Dandora, Kenya
September 2025 – January 2026

- Led a community digital literacy initiative from curriculum design to delivery, training 50+ residents (youth and adults) in foundational computer skills, internet safety, and basic programming – achieving an 80% course completion rate.
- Developed and maintained the organisation's website with a custom CMS interface that allows non-technical staff to update content (news, events, resources) without coding, reducing update turnaround from days to hours.
- Created interactive JavaScript learning modules using React and a localised code editor, enabling students to practice coding directly in the browser; integrated a progress tracking system that visualised completion rates and assessment scores.

Machine Learning Engineer | Matakiri Tumaini Trust

(Community Development Organisation)

- Built time series forecasting models (Python, Scikit-learn, Pandas) to predict resource needs (food, supplies, volunteer hours) for community projects, achieving 85% accuracy and helping the trust reduce waste by 20% through better inventory planning.
- Deployed an NLP sentiment analysis pipeline using Hugging Face Transformers (BERT-based model) on community feedback comments from WhatsApp groups and SMS surveys, automatically categorising responses as positive, neutral, or negative to prioritise follow-up actions.
- Created interactive dashboards with Matplotlib and Seaborn for non-technical stakeholders, visualising project impact metrics (e.g., number of beneficiaries served per month, resource utilisation) – leading to more data-driven donor reports.
- Engineered a data collection system using Google Forms integrated with Python scripts to clean, validate, and store feedback in a PostgreSQL database, reducing manual data entry errors by 90%.

Data Analyst Attache | Rongo University (Migori, Kenya)

June 2023 – August 2023

- Analysed 1,000+ academic datasets (student enrolment, exam performance, library usage) using SQL and Excel, identifying trends that informed departmental budgeting and improved decision-making accuracy by 20%.
- Executed complex SQL queries on a database of 100,000+ student records, optimising joins and indexing to reduce average query retrieval time by 30%, which sped up reporting for faculty and administration.
- Developed 15 interactive visualisations using Tableau and Power BI, presenting insights on graduation rates, course pass percentages, and resource allocation to the university senate.
- Standardised data collection protocols across five departments, creating validation rules and data entry templates that increased data integrity from 85% to 98% within two months.

Marketing Analyst | Lenku Marketing Firm (Remote)

January 2023 – March 2023

- Conducted market research and customer segmentation using Google Analytics and survey data, identifying two high-value customer personas that led to a 25% increase in email campaign open rates.
- Optimised SEO for client websites by performing keyword research, on-page optimisation (meta tags, heading structure), and backlink analysis, resulting in a 40% increase in organic search traffic for a key e-commerce client.
- Analysed campaign performance metrics (CTR, conversion rate, CPA) and provided weekly reports with actionable recommendations, helping the firm reallocate budget to top-performing channels and reduce cost per lead by 15%.

Web Designer | Timnjiz Business Center (Nairobi, Kenya)

June 2024 – December 2024

- Designed and developed responsive websites for the company using HTML5, CSS3 and WordPress, resulting in a 50% average increase in online inquiries.
- Improved UI/UX by conducting usability testing and implementing client feedback across the sites.
- Ensured cross-browser compatibility and mobile responsiveness, achieving consistent rendering on Chrome, Firefox, Safari, and Edge, and passing Google's mobile-friendly test for all projects.
- Provided ongoing maintenance and training to business owner, empowering him to update basic content (hours, prices, special offers) without developer assistance.

Software Engineering Projects

Xgene Labs – Molecular Diagnostics Platform
Full-stack e-commerce solution with M-Pesa mobile payments for a Kenyan diagnostics company.

- **Tech Stack:** React (Vite), Node.js/Express, MongoDB, Tailwind CSS, M-Pesa Daraja API, JWT, Cloudinary
- **Key Contributions:**
    - Built a production-grade online store for genetic testing kits, reagents, and medical equipment.

  - Integrated M-Pesa STK Push – handled payment callbacks, transaction verification, and order status updates.
  - Designed an admin dashboard with inventory management, order tracking, and sales analytics.
  - Optimised MongoDB queries for product search and filtering, reducing response time by 40%.
- **Outcome:** To enable the client to accept mobile payments for the first time, increasing online sales.

SomaNasi – AI-Powered Learning Management System
Collaborative LMS with integrated AI tutoring assistant using Hugging Face NLP.

- **Tech Stack:** React, Node.js/Express, MongoDB, Hugging Face Transformers (DistilBERT), Tailwind CSS, Socket.IO
- **Key Contributions:**
  - Developed a full LMS with course creation.
  - Fine-tuned a DistilBERT model on educational Q&A data to power a real-time AI tutor that answers student questions and provides explanations.
  - Built a responsive chat interface with typing indicators and conversation history.
  - Implemented role-based dashboards (Admin, Instructor, Student) with granular permissions.
- **Outcome:** increased student engagement by 45% through 24/7 AI assistance.

TinyWow Clone – Document & Image Processing Platform (Dockerized)
Distributed system for asynchronous PDF and image processing.

- **Tech Stack:** Flask, React, PostgreSQL, Redis, Celery, Docker, Docker Compose, Nginx, PyPDF2, Pillow, Tailwind CSS
- **Key Contributions:**
  - Architected a microservices-style system where file processing tasks are queued via Celery and executed asynchronously.
  - Implemented PDF compression, splitting, merging, password protection, and image format conversion (PNG/JPEG/WebP).

    - Containerised all services (Flask API, React frontend, Redis, PostgreSQL, Celery worker) using Docker Compose.
    - Added real-time job status polling with a drag-and-drop file upload interface.
- **Outcome:** Handles 10+ concurrent file operations with average processing time under 5 seconds per file; used internally by a small media agency.

Siprosa Foundation Website
Modern, responsive website for an educational non-profit in Kenya.

- **Tech Stack:** React 18, Vite, Tailwind CSS, Framer Motion, React Router, Lucide React, React Hook Form, EmailJS
- **Key Contributions:**
    - Built a multi-page, SEO-optimised site highlighting the foundation's "Futures Green Schools" model.
    - Implemented smooth scroll animations, fade-in effects, and interactive cards using Framer Motion.
    - Created a functional contact form with validation and google integration for automatic message delivery.
    - Ensured 100% mobile responsiveness and 95+ Lighthouse performance score.

Kenyan Real Estate Platform
Full-stack marketplace with real-time chat, M-Pesa subscriptions, and role-based dashboards.

- **Tech Stack:** React, Node.js/Express, MongoDB, Socket.IO, Google Maps API, M-Pesa API, JWT, Cloudinary
- **Key Contributions:**
    - Built property listing CRUD with image/video uploads, search/filter by location, price, and type.
    - Integrated Google Maps API to display property locations and nearby amenities.
    - Implemented real-time messaging between landlords and clients using Socket.IO, with message persistence in MongoDB.
    - Added M-Pesa subscription payments for landlords (monthly listing fees) with automatic expiry handling.

- Created three dashboards: Client (favourites, saved searches), Landlord (manage listings, respond to messages), Admin (manage users, moderate content).

- **Outcome:** Streamlined property discovery and communication, reducing average rental closing time by 2 weeks.

Dantra Limited – FMCG Distributor Website
Professional B2B showcase for a leading FMCG distributor in Kenya.

- **Tech Stack:** React, Tailwind CSS, Framer Motion, shadcn/ui, Vite, Vercel
- **Key Contributions:**
  - Designed a mobile-first, animated landing page with hero section, product categories, and brand partners.
  - Integrated WhatsApp API for instant partnership inquiries and click-to-call for phone numbers.
  - Built a reusable component library (cards, modals, buttons) for consistency and future scalability.
- **Outcome:** Provided a modern digital presence.

Sonar Rock vs Mine – ML Classification UI
Interactive web app for sonar signature classification (mines vs rocks).

- **Tech Stack:** React, Tailwind CSS, JavaScript, (mock ML API ready for real backend integration)
- **Key Contributions:**
  - Built a real-time classification interface that accepts CSV strings of 60 normalised sonar features.
  - Implemented input validation (exactly 60 numbers, comma-separated, range 0-1) with user-friendly error messages.
  - Added confidence scoring and colour-coded results (red for mine, green for rock).
  - Provided three pre-loaded sample datasets (rock, mine, edge case) for demonstration.
  - Designed a clean, responsive UI with one-click clipboard copy for sample data.

- **Outcome:** Demonstrates practical ML integration in a maritime safety context; ready to connect to a real model API.

Gym Sable One – Fitness Landing Page
Sleek, animated gym website showcasing programs and memberships.

- **Tech Stack:** React, Tailwind CSS, Framer Motion, Lucide Icons, Vite, Vercel
- **Key Contributions:**
  - Developed hero section with call-to-action, program cards (strength, cardio, yoga), and pricing tables.
  - Added scroll-triggered animations and hover effects for an engaging user experience.
  - Fully responsive layout that adapts to mobile, tablet, and desktop views.
- **Outcome:** (simulated demo).

The Eleventh Hour – Coffee Shop Demo (UK)
Demo website for a UK-based coffee shop – menu, gallery, location, contact.

- **Tech Stack:** React, Tailwind CSS, Framer Motion, Lucide Icons, Vite, Vercel, Google Maps Embed API
- **Key Contributions:**
  - Built a categorised menu (coffee, tea, food) with prices and descriptions.
  - Implemented a responsive photo gallery showcasing shop interior, exterior, and product shots.
  - Embedded Google Maps for location and opening hours; added a contact form with validation.
  - Designed a warm, inviting colour palette and smooth page transitions.
- **Outcome:** Served as a portfolio piece demonstrating ability to create brand-focused lifestyle websites.

Matakiri Client Portal – Community Organization Website
Client-facing site for Matakiri (community trust) – mission, projects, impact stories.

- **Tech Stack:** React, React Router, Tailwind CSS, Framer Motion, Lucide Icons, Vite, Vercel
- **Key Contributions:**
    - Structured a multi-page site (Home, About, Projects, Get Involved, Contact) with consistent layout.
    - Created reusable components (ProjectCard, StoryCard, CTAButton) for easy content updates.
    - Added a "Success Stories" section with beneficiary testimonials and photos.
    - Integrated a functional contact form with validation and email forwarding.
- **Outcome:** Enhanced the organisation's online presence.

Matakiri Admin Dashboard – CMS
Secure admin panel for managing Matakiri website content (projects, stories, submissions).

- **Tech Stack:** React, Tailwind CSS, Chart.js, React Router, Framer Motion, Vite, Vercel, Mock API (JSON Server)
- **Key Contributions:**
    - Built full CRUD interfaces for projects (title, description, image, status), success stories, and team members.
    - Added an analytics dashboard with charts (Chart.js) showing form submission trends and page views.
    - Implemented a mock authentication system (JWT-style) with protected routes (ready for real backend).
    - Designed a mobile-responsive admin UI with confirmation modals, toast notifications, and search/filter.
- **Outcome:** Empowered non-technical staff to update website content instantly, reducing developer intervention by 90%.

Machine Learning & Data Science Projects (Elaborated)

FactChecker API
RESTful API that verifies the truthfulness of factual statements using machine learning.

- **Tech Stack:** Python, Flask, Scikit-learn, Pandas, LIAR dataset, TF-IDF, Logistic Regression
- **Key Contributions:**
    - Trained a text classification model on 12k labelled political statements (true, false, half-true, etc.).
    - Exposed a REST endpoint that accepts a statement and returns a probability score and label.
    - Deployed as a lightweight API using Flask and Gunicorn, with input sanitisation and rate limiting.
- **Outcome:** Achieved 72% accuracy on the test set; used as a prototype for a fact-checking Chrome extension.

BERT Misinformation Detector
Enhanced fake news detection using a fine-tuned BERT model.

- **Tech Stack:** Hugging Face Transformers, PyTorch, Pandas, scikit-learn, Kaggle Fake News dataset
- **Key Contributions:**
    - Fine-tuned bert-base-uncased on 20k labelled news articles (real vs fake).
    - Implemented data preprocessing (tokenisation, attention masks) and training loop with early stopping.
    - Achieved 94% validation accuracy, outperforming baseline logistic regression (78%).
    - Saved and exported the model to Hugging Face Hub for easy reuse.
- **Outcome:** Demonstrated advanced NLP skills and ability to work with transformer models.

Heart Disease Predictor
Machine learning model predicting cardiovascular risk based on health indicators.

- **Tech Stack:** Python, Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, XGBoost
- **Key Contributions:**
    - Analysed the UCI Heart Disease dataset (303 samples, 14 features).

    - Performed EDA, handled missing values, normalised features, and addressed class imbalance using SMOTE.
    - Compared Logistic Regression, Random Forest, and XGBoost – XGBoost achieved 89% accuracy.
    - Created a confusion matrix and ROC curve visualisation for stakeholder presentation.
- **Outcome:** Provided a proof-of-concept for a potential clinical decision support tool.

Crop Recommendation System
AI-driven model suggesting optimal crops based on soil and weather conditions.

- **Tech Stack:** Python, Scikit-learn, Pandas, NumPy, Random Forest Classifier, Streamlit
- **Key Contributions:**
    - Trained on 2,200 samples with features: nitrogen, phosphorus, potassium, temperature, humidity, pH, rainfall.
    - Random Forest achieved 99% accuracy on test data.
    - Built an interactive web app using Streamlit where farmers can input soil parameters and receive crop recommendations.
    - Deployed on Streamlit Cloud for free access.
- **Outcome:** Can help small-scale farmers make data-driven planting decisions.

Sonar Rock vs Mine Model
Classification model for sonar signal differentiation (rocks vs mines).

- **Tech Stack:** Python, Scikit-learn, Pandas, NumPy, Logistic Regression, KNN, Jupyter Notebook
- **Key Contributions:**
    - Worked with the UCI Sonar dataset (208 samples, 60 numerical features).
    - Performed feature scaling, train-test split, and hyperparameter tuning via GridSearchCV.
    - Logistic Regression achieved 86% accuracy; KNN with k=5 achieved 90%.
    - Visualised feature importance and decision boundaries using PCA.
- **Outcome:** Served as the backend logic for the Sonar Rock vs Mine UI project.

Walmart Sales & Ultra Marathon Analysis
Exploratory data analysis (EDA) projects demonstrating data cleaning, visualisation, and insight generation.

- **Tech Stack:** Python, Pandas, NumPy, Matplotlib, Seaborn, Plotly
- **Key Contributions (Walmart):** Analysed 45 stores' weekly sales over 3 years, identified holiday effects, created time-series plots, and built a simple linear regression baseline.
- **Key Contributions (Ultra Marathon):** Cleaned and analysed a dataset of 20k ultra-marathon race results, visualised finish times by age group, gender, and race distance.
- **Outcome:** Developed strong data storytelling skills and proficiency with Pandas/Seaborn.

Certifications

- SQL Basics – HackerRank
- Certificate in Computer Studies & Microsoft Packages – GIBS College

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Education

Bachelor of Science in Informatics | Rongo University, Kenya (Graduated 2024)
Coursework: Data Mining, Machine Learning, Statistical Analysis, Database Systems, Algorithms

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Languages

- **English** – Fluent (Professional working proficiency)
- **Swahili** – Fluent (Native)
- **French** – Basic